

SKILL : WEEK 5

Task -1 :

To Change Directories, Validate Paths, Update Working Directory, Handle Errors, Maintain Previous Directory, Test Navigation

CODE:

```
#include <stdio.h>
#include <unistd.h>
#include <string.h>

int main()
{
    char cwd[100], old[100];

    getcwd(cwd, sizeof(cwd));
    strcpy(old, cwd);

    printf("Current: %s\n", cwd);

    if (chdir("/tmp") == 0)
        printf("Changed to: /tmp\n");
    else
        perror("chdir");

    getcwd(cwd, sizeof(cwd));
    printf("Current: %s\n", cwd);

    if (chdir(old) == 0)
        printf("Returned to previous directory: %s\n", old);
    else
        perror("chdir");

    return 0;
}
```

OUTPUT:

```
b.deepak@BDeepaks-MacBook-Air ~ % nano Navigation.c
b.deepak@BDeepaks-MacBook-Air ~ % gcc Navigation.c -o Navigation
b.deepak@BDeepaks-MacBook-Air ~ % ./Navigation
Current: /Users/b.deepak
Changed to: /tmp
Current: /private/tmp
Returned to previous directory: /Users/b.deepak
b.deepak@BDeepaks-MacBook-Air ~ %
```

Task-2:

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic

CODE:

```
#include <stdio.h>
#include <string.h>
#include <unistd.h>

void pwd()
{
    char dir[100];
    getcwd(dir, sizeof(dir));
    printf("Current directory: %s\n", dir);
}

void cd(char *path)
{
    if (chdir(path) != 0)
        perror("cd");
}

void exit_shell()
{
    printf("Shell terminated.\n");
}

int main()
{
    char command[100];

    char *commands[] = {"pwd", "cd", "exit"};

    printf("Mini Shell\n");

    while (1)
    {
        printf("shell> ");
        fgets(command, sizeof(command), stdin);
        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "pwd") == 0)
        {
            pwd();
        }
        else if (strncmp(command, "cd ", 3) == 0)
        {
            cd(command + 3);
        }
        else if (strcmp(command, "exit") == 0)
        {
            exit_shell();
            break;
        }
        else
        {
            printf("Invalid command.\n");
        }
    }

    return 0;
}
```

OUTPUT:

```
[b.deepak@BDeepaks-MacBook-Air ~ % nano Builtins.c
[b.deepak@BDeepaks-MacBook-Air ~ % gcc Builtins.c -o Builtins
[b.deepak@BDeepaks-MacBook-Air ~ % ./Builtins
Mini Shell
shell> pwd
Current directory: /Users/b.deepak
shell> cd /tmp
shell> pwd
Current directory: /private/tmp
shell> hello
Invalid command.
shell> exit
Shell terminated.
```